MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
TECHNICAL SERVICES DIVISION — ROTATING EQUIPMENT ENGINEERING

ENGINEERING TECHNICAL STANDARD: PUMP NPSH & CAVITATION MARGIN CRITERIA
DOC NO: **MRPL-ENG-STD-402** | REV: 03 | REFERENCE: API 610 / ISO 13709

1. GOVERNING NPSH FORMULATION

Net Positive Suction Head Available (NPSHa) represents the total suction head available at the pump suction centerline over and above the vapor pressure of the pumped liquid at operating temperature. The mandatory calculation is:

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NPSHa = h_sp + h_s - h_f - h_vp
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Where:
• **h_sp** = Absolute pressure on the liquid surface in suction vessel = (P_suct_abs) / (ρ · g) [meters]
• **h_s** = Static liquid level above pump impeller centerline (positive if above, negative if below) [meters]
• **h_f** = Total piping friction and dynamic head losses (suction pipe, strainer, valves, fittings) [meters]
• **h_vp** = Vapor pressure of hydrocarbon mixture at maximum design pumping temperature = (P_vap) / (ρ · g) [meters]

2. MANDATORY CAVITATION SAFETY MARGIN RULE

Per MRPL Engineering Standard and API 610 §6.1.8:
NPSHa ≥ **NPSHr + 1.0 meter** (or **1.20 × NPSHr**, whichever margin is GREATER).
For hydrocarbons with specific gravity < 0.80 operating near boiling point (bubble point), an additional 0.5 m vapor suppression contingency is enforced.

3. VERIFIED DESIGN CALCULATION: FEED PUMP P-101A (NAPHTHA STABILIZER)

Parameter	Design Value	Converted Head	Notes
Operating Temperature (T)	65.0 °C (338.15 K)	—	Maximum summer ambient scenario
Liquid Density (ρ)	740.0 kg/m³	—	Light Naphtha fraction
Suction Vessel Pressure (P_suct)	1.80 bar(g) = 2.813 bar(a)	38.74 m	h_sp = (2.813 × 10■) / (740 × 9.81)
Static Liquid Height (h_s)	+4.20 meters	+4.20 m	Minimum operational low level (LLLL)
Suction Piping Friction (h_f)	0.85 bar loss equivalent	-1.17 m	Includes 100-mesh suction strainer
Hydrocarbon Vapor Pressure (P_vp)	0.420 bar(a)	-5.78 m	h_vp = (0.420 × 10■) / (740 × 9.81)
Calculated NPSHa	—	35.99 m	NPSHa = 38.74 + 4.20 - 1.17 - 5.78
Vendor Certified NPSHr	3.20 meters	3.20 m	At rated flow 145 m³/h (1480 RPM)
Design Margin (NPSHa - NPSHr)	—	+32.79 m	COMPLIES (Excess exceeds API 610 margin)